ARYAN AMIT TIWARI

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Developer shipping applied AI and distributed systems. I build the backend infrastructure that powers autonomous agent workflows, quantitative data pipelines, and Web3 protocols, focusing on fast execution and clean, scalable code.

EXPERIENCE

VeLyra Labs(Stealth Startup)
Gen AI Intern — 07/2026 - Present

- Architected scalable system topologies, translating complex business requirements into backend infrastructures that cleanly isolate heavy, latency-sensitive AI workloads from high throughput transactional services.
- Developed and enhanced generative AI capabilities by building intelligent application workflows, integrating organizational data, and ensuring secure, reliable, and scalable system behavior.

Wayspire (Microsoft & Tryst, IIT-Delhi)
A.I. Intern (Remote) — 02/2024 - 03/2024

- Incorporated and tested scalable NLP modules for intent classification, achieving accuracy by systematically analyzing model performance and refining algorithms.
- Translated business requirements into technical specifications to deliver features and support implementation for the capstone project.

PROJECTS

StakePort

- A blockchain platform for fractional ownership of high value assets using Web3 tokenization.
- Dividing each asset into digital tokens that represent ownership shares, enabling easy buying, selling, and trading of asset fractions.
- Leveraging a secure and transparent blockchain network to facilitate transactions.
- Lowering investment barriers, improving liquidity, and access to valuable assets while ensuring security and transparency.
- Implementing smart contracts to handle ownership transfers.

EcoLoop

- An autonomous multi-agent building controller that dynamic optimizes EnergyPlus HVAC simulations to reduce energy consumption while maintaining thermal comfort.
- Engineered a decoupled architecture featuring an LLM cognitive engine for strategy, a deterministic supervisor for setpoint calculation, and a Guardian safety shield to enforce physical thermodynamic constraints.
- Built fault-tolerant data pipelines with self-repair mechanisms for simulation stability and an interactive dashboard to monitor real-time power and comfort metrics.
- Validated performance across extreme climate profiles (Chicago & Delhi) using a rigorous ablation ladder against traditional controllers.

MediVault

- A decentralized unified health ledger that grants patients absolute sovereign ownership of their medical history using Web3 architecture.
- Leveraging a hybrid on/off-chain architecture, bridging a React and Node.js ecosystem with the Ethereum Sepolia network for secure data interoperability
- Eliminating centralized data silos and unauthorized breaches, while ensuring instant, permissioned access to vital records for verified medical practitioners.
- Solidity smart contracts via Ethers.js to enforce granular access control.

Recon

- Building a fully autonomous, multi-agent AI system to simulate a quantitative hedge fund's research and execution pipeline.
- Developing a specialized "crew" of AI agents that scrape financial data, debate risk, and formulate structured trading strategies using orchestration frameworks.
- Integrating high-speed data pipelines, caching, and local NLP models to condense market noise and execute analysis.
- Engineering an instant backtesting loop to validate generated strategies against historical data.

SKILLS

- Programming: Python, TypeScript, JavaScript, C++, C, SQL, Java
- AI & ML: LLM's, RAG, Autonomous Agents, Prompt Engineering, Fine Tuning, PyTorch, Deep Learning
- AI Frameworks: LangGraph, n8n LangChain, Hugging Face
- Databases: MongoDB, VectorDB, SQLite, PostgreSQL
- Cloud & DevOps: AWS, Docker, Azure, Git/GitHub, CI/CD
- Backend & APIs: FastAPI, Node.js, REST/GraphQL APIs, Web3/Blockchain, Smart Contracts, Cryptography

EDUCATION

VIT Chennai — **2023 - Present**

- BTech in CSE(AI & ML)

IIT Madras — **2023 - Present**

- BS in Data Science

DPS Sushant Lok, Gurgaon

- Schooling

CERTIFICATIONS

NPTEL : EBusiness(IIT-K)
Business Fundamentals(IIT-B)
IBM: Gen AI
Finlatics: Business Analyst

INTERESTS

- Open Source Contributions
- Decentralized Systems
- Quantitative Finance & Algorithmic Trading